Training a (RAG) retrieval model

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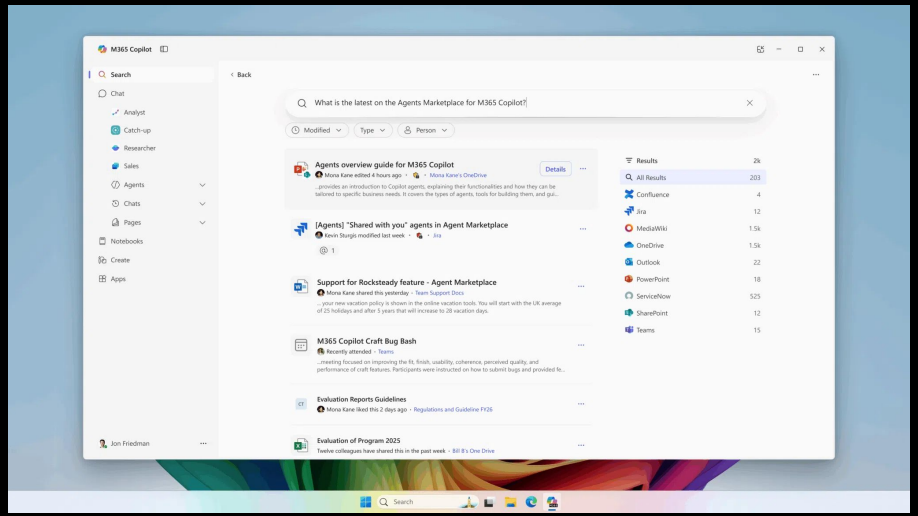


ChatGPT with company knowledge

Summarize customer feedback and scope the next mobile release

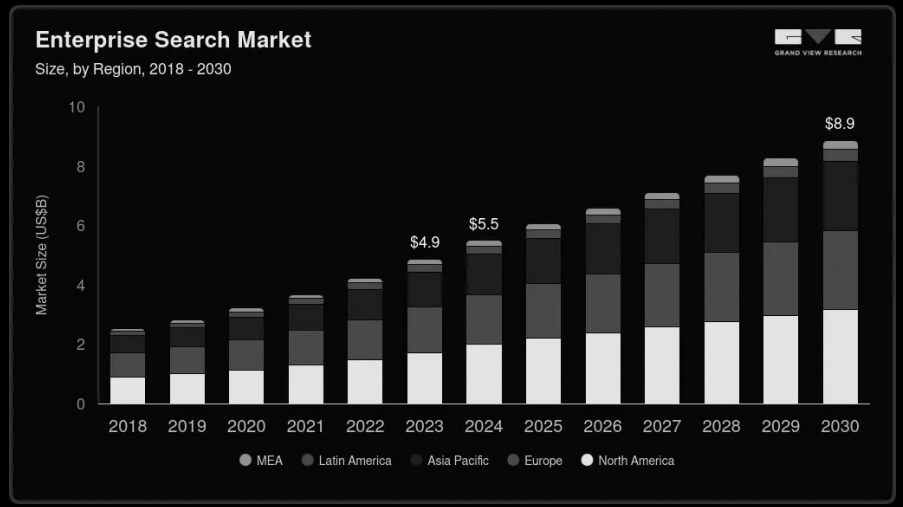
+ Company knowledge 3

Overview

Enterprise search market is ~$6B and growing to ~$9B by 2030. Every company needs system for employees to find relevant enterprise docs.

Goal: Build a high quality retrieval system for search engines and LLMs (RAG) over NVIDIA synthetic corpus

Document Index
Chunking
Embedding
RAG (Supervised ML model) fusing lexical + semantic



What is our data?

We have a synthetic dataset created by Nvidia. It has a list of:

1. User queries - questions asked by users
2. Document (Index) Properties - chunks, topics, key concepts, tags, etc
3. Passage - this was the summarized response from returned doc(s), chunk(s)
4. Evaluation (label) - our target of returning a good or bad result

User query	Relevant_doc_Id	Passage	Quality_Score
What is the role of apples-to-apples comparison...	5	Apples-to-Apples is important when comparing….	9.0

https://huggingface.co/datasets/nvidia/Retrieval-Synthetic-NVDocs-v1

What will we build?

To build a system that can return the most relevant answer to a user question, we will need to build:

1. Document Index - A collection of all the documents and properties a user can search over
2. Chunking method - Parts or “chunks” of a single document used to match on the user query
3. Ranking Model - Supervised (Decision Tree, XGBoost, etc) model that returns the most returns relevant passage, document
4. Evaluation & Explainability - NDCG, Precision@n, Recall@n & SHAP values

Timeline / Roadmap

Goal	*ETA*
EDA, Document Index	3/25
Data cleanup, feature engineering, split	4/1
Training models, Evaluation	4/8
Summarize, Visualization - Presentation	4/15
Submit	4/22